



OPEN Remote sensing and GIS-based modelling of land use dynamics and urban flood risk in Lagos megacity for future flood mitigation

Opeyemi Aniramu^{1✉}, Olamide Iyanda¹ & Oluwagbenga Orimoogunje²

This study investigates changes in land use and land cover (LULC) and their impact on flood risk in Lagos Megacity using an integrated geospatial approach. Multi-temporal Landsat TM1984, ETM + 2002, OLI 2023, and Sentinel-2 images were used for LULC classification. Flood depth mapping was conducted using terrain models and hydro-climatic datasets to assess surface inundation dynamics. Rainfall-runoff modelling and flood hydrograph simulations were performed using HEC-HMS 4.12, while Markov Chain Model predicted future LULC dynamics and flood receptors by 2050 within the megacity. Results revealed significant urbanization, with light vegetation (18.87%) and built-up areas (37.68%) expanded between 1984 and 2023. Meanwhile, forests (−31.55%) and waterbodies (−11.26%) declined rapidly, reducing the natural flood buffering capacity of Lagos Megacity. Flood impact analysis revealed that 46, 018.18 ha were affected within 12–24 h; 125, 218.43 ha over 5–7 days; and 211, 230.22 ha were severely affected for up to 30 days. By the year 2050, extreme flooding will significantly impact built-up area, totaling 236,810.7 ha (66.20%), while natural flood buffers including forest (1.35%), light vegetation (7.64%), and waterbodies (10.86%) will decline drastically over the predicted year, indicating noticeable environmental changes and high flood vulnerability in the future. The spatial modelling highlighted the need for robust disaster reduction framework, emphasizing the interaction between land use dynamics and hydro-climatic responses in flood-prone areas of Lagos. The study is timely and crucial for informing sustainable land use frameworks and the implementation of future flood mitigations in fast-growing African cities such as Lagos.

Keywords Land use and land cover, Urban flood risk, Markov chain model, Urban growth dynamics, Flood mitigation, Lagos megacity

The human environment is undeniably subject to development and land utilization policy remains at the forefront of sustainable urban growth¹. The use of land, land transformation, and land cover are supposed to be done according to sustainable development plans in order to reduce flood cases and other natural hazard risk². Natural disasters in most developing nations have been aggravated by the emergence of ecological risks related to the land use dynamics and incessant urbanization³. Wetlands, water bodies, and waterlogged areas are some of the ecologically sensitive areas which need proper protection.

Land use indicates the interaction between the human, their habitats, and socio-economic activities⁴. Previous studies on land-use dynamics indicated that the existing trends of land use patterns has led to degradation, which endangers the future sustainability and the urban resources^{5–7}. The United Nations-Habitat World Report⁸ identifies the interconnectedness between urbanization, climate effects, and city development that plague urban population to increasing flood risk. In Asia and Africa, urban population is projected to be 68 percent of the global population in 2050, with Nigeria due to have almost 295 million population by that time, making it the urban giant of the African continent^{9,10}.

Lagos state, which is the economic powerhouse of Nigeria, has about 30 percent contribution to the national GDP; yet continued to face rapid urbanization and increasing flood risks¹¹. Urban flood vulnerability is triggered by the population growth of the city¹², coupled with poor land-use planning, inadequate drainage facilities,

¹Institute of Ecology and Environmental Studies, Obafemi Awolowo University, Ile-Ife, Nigeria. ²Department of Geography, Obafemi Awolowo University, Ile-Ife, Nigeria. ✉email: samuelaniramu@gmail.com

climate change impacts, and encroachment into the floodplains¹³. The inadequacies in land use and land cover (LULC) roadmap due to high rate of demographic and spatial expansion, have caused a strain on the land resources¹⁴, resulting in unsustainable LULC patterns that disturb the megacity hydrological network and flood buffering capacity¹⁵. Additionally, Lagos impervious area has increased direct discharge effect of runoff during acute floods, extending the time needed by the floodwater to move to the drainage systems¹⁶. Lagos is characterized by frequent flooding caused by natural and anthropogenic activities, including unsustainable urbanization and poor land cover framework^{17,18}. More recently, high floods are caused by extreme and shorter length rainfall events, while land area and drainage networks were put under strain¹⁹.

To plan urban areas, it is important that urban flood is monitored; such that land use dynamics of spatial heterogeneity is evaluated. However, this assessment has long been hampered by data constraints and technical capacity, nevertheless, GIS and remote sensing have become reliable and effective in flood hazard assessment and prediction^{20–25}. The LULC detection analysis enhances the decisions of urban planners to determine areas where development can be executed, monitored, or halted²⁶. The growth of urban areas, deforestation, and wetland destruction can be monitored using satellite images²⁷. The application of supervised classification based on Maximum Likelihood Classifier (MLC) has been introduced to map land-use dynamics and flood buffer reduction^{28–30}.

The development of flood hazard mapping has progressed in terms of weighted overlay analysis with the inclusion of Multiple-Criteria Decision-Analysis (MCDA) that are used to evaluate the effects of the population growth, urban development, physical configuration, and climatic parameters^{31–36}. Landscape factors like slope, soil types, drainage patterns, and terrain curvature are physical factors that have direct impacts on flood vulnerability³⁷. Analytical Hierarchy Process develops a methodological prioritization of the flood components in order to obtain the preferred flood solutions^{38–41}. Nonetheless, a standalone model has drawback the ability to predict future changes in LULC and related effects of urban flooding. This limitation necessitate a combination of flood risk spatial model of land use dynamics, incorporating GIS-based algorithms of Cellular Automata-Markov, to forecast the future pattern of LULC prediction and flood receptors, a novel approach of the present study. The Hydrologic Modeling System (HEC-HMS) offers the benefits of characterizing urban flood zones through the simulated catchment run-off during rainfall events using hydro-climatic data^{42–44}. The alterations in LULC could be then studied to comprehend their effect on surface runoff, infiltration, and flood mitigation^{45–47}, which can be used to guide mitigation efforts in Lagos megacity and other global cities where similar characteristics exist.

The broadening of research on climate change, land-use resources, and ecological vulnerability is a global agenda. Regional peculiarities are often tied to rapid population growth, unplanned urbanization, poor implementation and governance especially in the global south⁴⁸. Lagos witnessed high flood vulnerability in the coastal and low-lying areas, reinforcing the unregulated urbanization and inadequate land-use systems^{49,50}. In fact, sea-level rise of about 3 m is expected by 2050 with severe rainfall, putting over 1.4 million inhabitants in direct flood pathway and urban resource collapse^{16,51}. There is a risk of flood affecting about 82 percent of the urban areas⁵². Such estimations highlight the necessity of tougher disaster and flood prevention model as one of the Sustainable Development Goals (SDGs) in Lagos megacity, Nigeria.

This study evaluates spatial patterns of urban flooding and land use dynamics in Lagos through remote sensing, HEC-HMS, and Markov Chain analysis, to improve future flood resilience. The land use classification was analyzed in 39 years (1984, 2002 and 2023) to identify variation of land cover. Surface inundation patterns were recorded through urban mapping of the flood depth using rainfall-runoff processes, whereas Markov Chain simulations indicated the future changes in urban flood receptors as a result of the further development of the land-use dynamics. This study is innovative in that it integrates dynamic land-use projections with flood simulation, moving beyond traditional static models in flood studies. It translates hydrological data into practical planning insights by combining flood metrics with land-use types, and links flood risk directly to urban growth pathways for policy-relevant resilience strategies. Furthermore, it presented an integrated strategy to discuss frequent flood issues in one of the most rapidly expanding African cities, where high flood risks, climate fluctuations, and unsustainable land use are common occurrence. Spatial and temporal modeling, which is based on remote sensing and GIS, allows identifying areas that are at risk of floods both now and in the future. The research is an advocate of evidence-based policies regarding flood mitigation, alignment of spatial resilience initiatives and land use planning, and a contributor to SDGs 11, 13, and 15, which seeks to reduce disaster risks and climate-resilient urban planning in coastal megacities such as Lagos, Nigeria⁵³.

Data and methods

This section presents the data acquisition and data processing methods for LULC analysis, flood depth mapping and prediction. Various methods used to analyze the accuracy and model validations of the studies are also included in this section.

The study area: Lagos state

The southwestern region of Nigeria (Fig. 1) lies Lagos State, a coastal megacity on the shallow floodplain of the Bight of Benin, and south of the Atlantic Ocean, on latitudes 6° 22' to 6° 42' N and longitudes 2° 42' to 4° 22' E. The population of Lagos has increased tremendously, doing 305,000 residents in 1950, to 9.1 million people in 2006 (National Population Commission, 2006) and is expected to be 21 million people by 2025. Majority of inhabitants live in water basins and the coastlines especially in the Lagos Metropolitan Area, whereas Epe, Ibeju-Lekki, Badagry, and Ikorodu are suburban⁵. Lagos State is one of the most congested urban cities in Africa, experiencing rapid urban transformation over the last few decades⁵⁴. The topography of the flat area is between 3 m in the south of Lagos to 73 m in the Ifako-Ijaiye area⁵⁵. The annual average rainfall is 1900 mm, while urban sprawls aggravate flooding in Surulere, Eti-osa, Lagos Island, and other LGAs (LASG, 2022). Children, women,

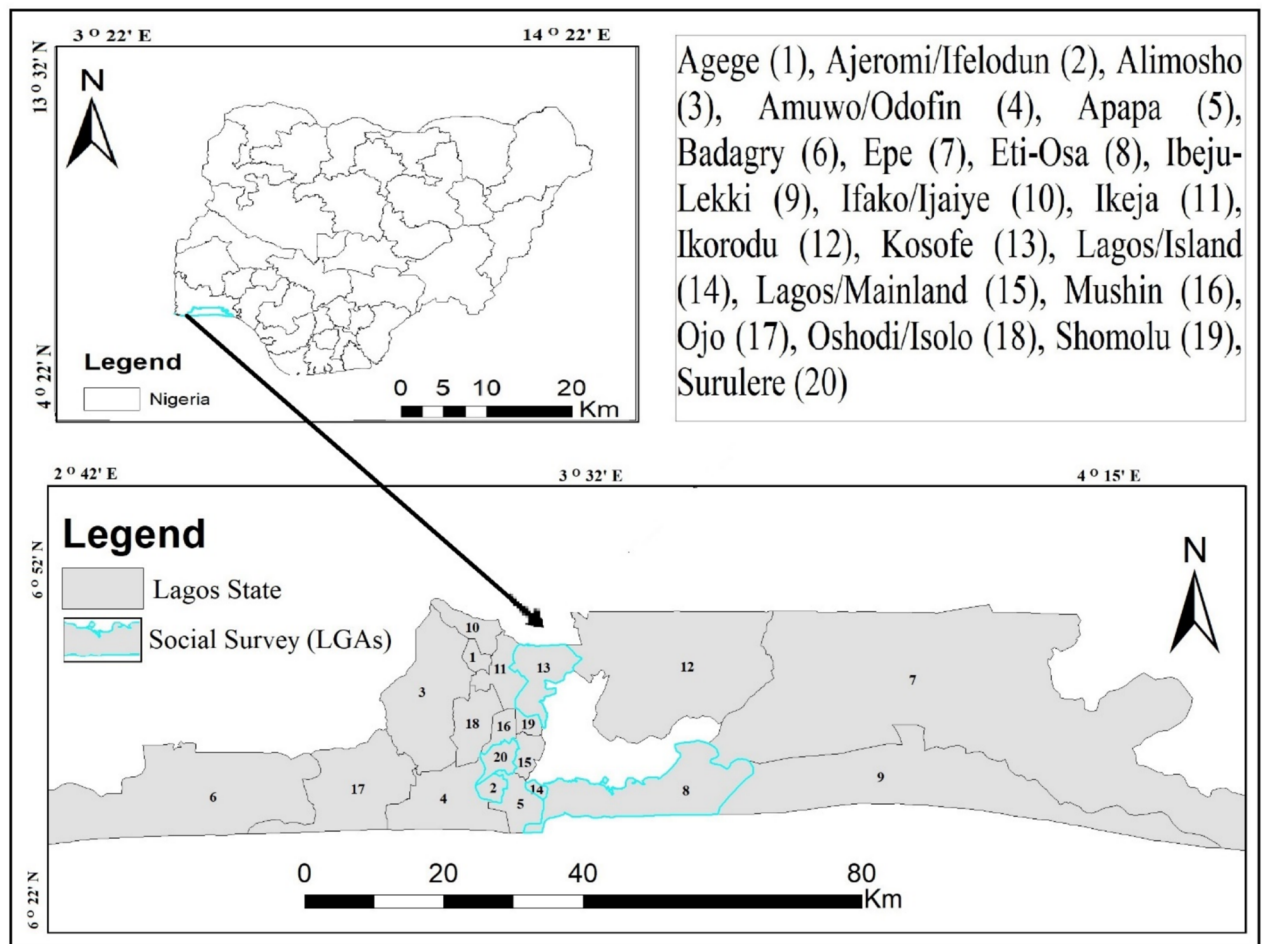


Fig. 1. Showing the map of the Lagos State, Nigeria.

and the physically challenged are some of the vulnerable populations⁵⁶. NiHSA⁵⁷ categorized significant Lagos regions into high-risk zones of flooding, showing a high need of present and future flood resilience.

Multi-temporal satellite data

The study employed the use of Landsat 4 Thematic Mapper (TM) of 1984, Landsat 7 Enhanced Thematic Mapper Plus (ETM+) of 2002, and Landsat 8 Operational Land Imager (OLI) of 2023 images. Landsat Level 1 datasets of the study area were acquired from the earth explorer portal (www.earthexplorer.usgs.gov/) of the USGS (United States Geological Survey (USGS)), with a minimum cloud opacity of 5% as recommended by USGS⁵⁸. Sentinel-2 Data, a high resolution data with a revisit cycle of five days support monitoring of earth's surface changes. Sentinel-2 products are approximately 110 × 110 km tiles in spatial cartographic projections (UTM/WGS84), available and accessible on the Copernicus Data Space Ecosystem. Flood mapping process utilized a spatial resolution of 10 m for visible, and near-infrared (VNIR) bands, and 20 m for red-edge and shortwave infrared (SWIR) bands to generate flood vulnerability maps⁵⁹. In addition, Digital Elevation Model (DEM) was employed to create topographic maps. DEM data are in raster format, where each pixel contains an elevation value representing the earth's surface configuration. The Shuttle Radar Topography Mission (SRTM) provides global and accurate elevation data at 30-m resolution, freely available on open access. SRTM-DEM is reprocessed data widely used in terrain mapping, surface-runoff generation, flood risk assessment, and hydrological modelling^{60–63}. DEMs are referenced to the vertical datum, which is the surface of zero elevation. The use of high and middle resolution satellite Images has been recognized as a reliable and accurate procedures in mapping, and detecting changes in the Land-use/ Land cover (LULC) over an epoch of time^{28,64,65}. Table 1, presents data information, their year and date of acquisition. The downloaded images had standard radiometric attributes⁶⁶. Therefore, they were processed geometrically with three infrared and three visible bands of spectral characteristics suitable for analyzing land cover change⁶⁷, flood inundation mapping and flood prediction⁶⁸. In the study, geographic coordinates of the observed flood impacts were collected the using GPS Model. These coordinates and their addresses were interpolated for more location-specific classification, enhancing the location-verification of land-use mapping process. The study combined primary and secondary data sources to achieve its objectives effectively, details of the data methodologies were provided in Table 1 and Fig. 2

S/N	Data description	Source/resolution	Relevance
1	Landsat: TM, ETM +, OLI	USGS/30 m by 30 m. (1984, 2002, and 2023)	For change detection analysis, quantifying the pattern of the urban terrain, and mapping of disaster zones in the Lagos State
	Sentinel-2	Europe's Copernicus/10 m, 20 m, and 60 m	
	SRTM-DEM	NASA/30 m	
2	Hydro-Climate Data: (a) Rainfall (b) Streamflow Discharge	(a) NiMET / 1984–2023 (b) O-ORBDA /1995–2018	To simulate flood trends, depth, and variability using peak data from the study area
3	Ground-Truthing (Garmin 64 s)	± 3 m	Verify vulnerable flood zone and spatial classification mapping
4	Administrative Map	LASG/ 2023	To guide boundary, drainage, and terrain delineation

Table 1. Description of materials and instruments used for the study.

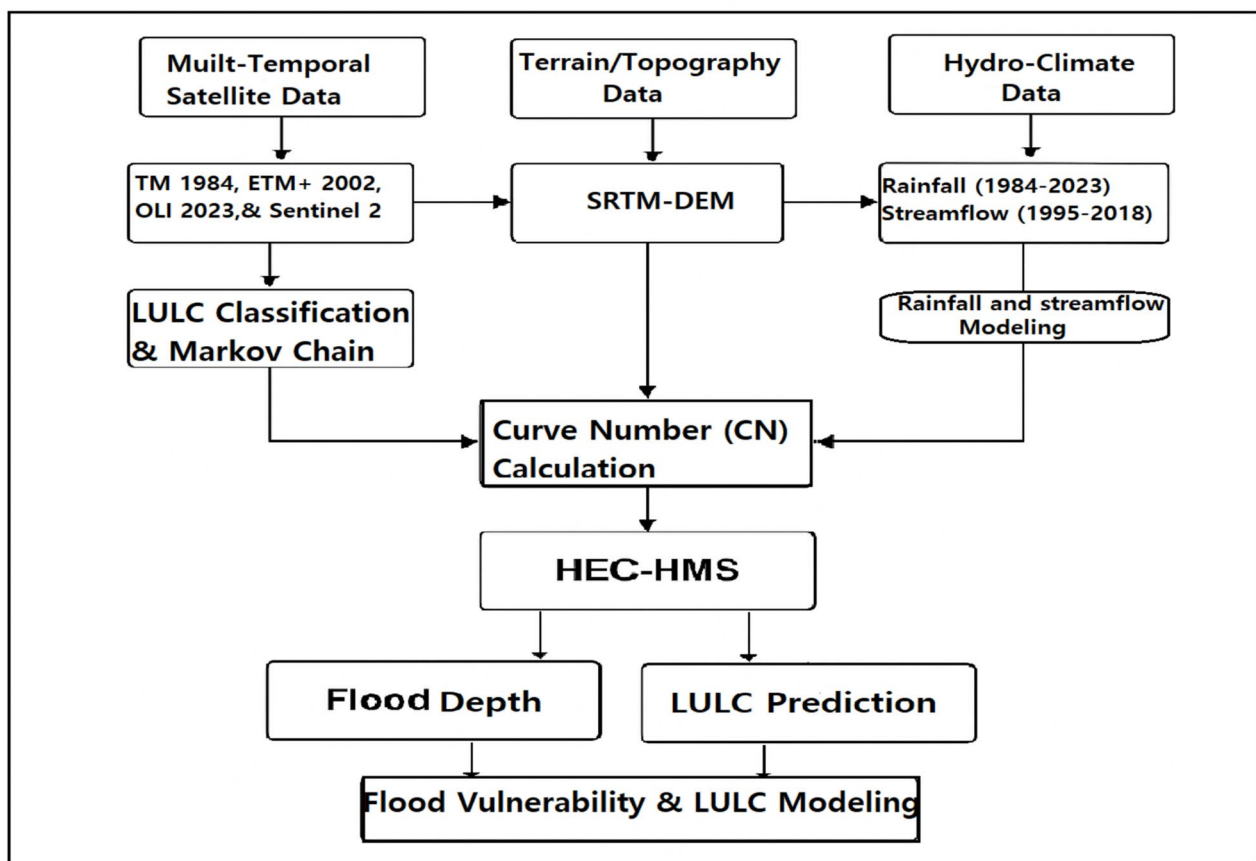


Fig. 2. Methodological flowchart diagram.

Multi-spectral imagery pre-processing

The Landsat images, according to Tewabe and Fentahun⁶⁹, satellite data registration was done to ensure images conformed to one another via the coordinate system of Lagos State, Nigeria. The geo-referenced images maintained the same coordinate system, WGS84 / UTM zone 31N. The field-based geographic coordinates obtained were recorded in longitude and latitude (x, y) and assigned to overlay different points on the geo-referenced map. Regularization of spatial resolution for Landsat images of TM 1984, ETM + 2002, and OLI 2023 in spatial resolutions of 28.5 m, 30 m and 30 m respectively were resampled. These variations needed to be harmonized in order to unify spatial dataset and enhance accuracy of spatial analysis tools. Image Mosaicking was conducted on sentinel and Landsat scene of Lagos State stretched across two different paths and rows. Downloaded Landsat images were then merged together into a single layer through the raster processing tools. This ensured the spatial assembly of image datasets to produce a spatially continuous single image covering the entire study area. Image clipping was performed to extract Lagos State from the larger mosaicked satellite image. This was done to facilitate easy and accurate spatial processing, focusing on the study area while reducing computational overhead. Clipping Lagos State's boundary for flood risk analysis, helps to focus on the region

of interest rather than the entire satellite scene. The study engaged in composite of selected bands from the satellite image based on the natural colour composite. The band combination of RGB revealed a real-world scenario, where green represents vegetation, blue represents waterbodies, red represents built-up and brown for bare soil. The appropriate band was selected for image composite based on the spectral characteristics, and spatial objectives to be achieved for the study. Precisely, an infrared color composites of 4-3-2 representing RGB was done to enhance the contrast, while extracting components of land cover characteristics in time and space.

Hydro-climate data processing

The study obtained rainfall and streamflow discharge data from ground-based station: Nigerian Meteorological Agency (NiMet), and Ogun-Oshun River Basin Development Authority, (O-ORBDA). Historical rainfall data, intensity and frequency over a period of forty years (1984–2023) from NiMET station, as well as streamflow discharge (1995–2018) from O-ORBDA were collected based on availability. Additionally, satellite data on Potential Evapotranspiration (PET) data for 2023 were acquired from flood depth simulations. PET data undergone initial standardization into a common format (NetCDF) to fit into ArcGIS environment. The study addressed missing or erroneous values using imputation methods. Climate data aggregation including daily, monthly, seasonal totals, and annual mean was conducted. The data cleaning identified the outliers and correct unrealistic values using statistical techniques, it also decompose data for variability studies. Data entry was done using Microsoft Excel 2019, to obtain statistical significance.

Changes in land use and land cover

The study followed the standard method described by Jensen⁷⁰, as re-modified by Chen et al.⁷¹ in the LULC classification scheme, aiming to understand urbanization trend through spatial flood dynamics of vegetated to non-vegetated responses in the study area. For each year, average of four (4) cloud-free Landsat/Sentinel-2 images were collected. Images were pre-processed and merged using mean composite techniques to produce a single representative LULC map per year, in readiness for supervised classification, change detection, and accuracy assessment. The supervised classification process based on the Anderson et al.⁷², categorize satellite images into LULC classes as shown in Table 2. The Maximum Likelihood Classifier was employed for supervised classification, selecting appropriate sample pixels in each image to represent distinct classes in the land cover types. These spectral signatures were developed from specified locations within the composite satellite imagery, encompassing six LULC types including built-up area, bareland, forest, light vegetation, wetland, and waterbodies.

Changes in area analysis were conducted for each LULC classes in unit of hectares per image year, measuring the spatial extent in relation to epoch rate change across LULC categories. The percentage of area change in the classified image was computed while Kappa coefficients were obtained by comparing labelled pixels with field objects to determine the accuracy of classification⁷³. Vector layers were digitized over the raster scene, representing various polygons overlapping different land use types. The classified images were depicted in maps for LULC change detection, with losses and gains represented in tables and charts to visualize changes over the study period. The change detection and trend analysis underwent rate and magnitude over the period of 1984, 2002, and 2023, using change detection method of two different images.

Magnitude = Magnitude of new year – previous year (1)

S/N	LULC types	Description
1	Built-up	These are populated areas within the permanent residence
2	Bareland	A part of a land appearing beside the surface with no vegetation cover
3	Dense Forest	The totality of land area covered with untapped vegetation for ≥ 100 years
4	Light vegetation	A typical secondary vegetated area including grasslands, pastoral scrubs
5	Wetland	A submerged land by water, permanent or seasonal waterlogged conditions
6	Waterbody	The area of land covered by river, spring or stream within the study area
Resolution (m) Wavelength (um) Landsat band		
30	0.43–0.45	Band 1 Visible Coastal Aerosol
30	0.45–0.51	Band 2 Visible Blue
30	0.53–0.59	Band 3 Visible Green
30	0.64–0.67	Band 4 Red
30	0.85–0.88	Band 5 Near-Infrared
30	1.57–1.65	Band 6 SWIR-1
30	2.11–2.29	Band 7 SWIR-2
30	0.50–0.68	Band 8 Panchromatic
30	1.36–1.38	Band 9 Cirrus

Table 2. Description of the satellite bands, land use and land covers classes. Source: Anderson et al.⁷², USGS⁵⁸.

The study followed the methodology of Shiferaw and Singh (2011), in assessing the change detection analysis for both LULC classes and annual changes. However, the percentage changes in the LULC classes were conducted using the rate of change formula expressed as:

$$\text{Percentage change} = \frac{\text{Magnitude of Change}}{\text{Base Year}} \times 100 \tag{2}$$

where magnitude represent the land area measured in hectares of a particular LULC classes, and base year is the images considered.

Similarly, the study conducted annual rate of change for LULC classes, the change of LULC trend was divided by 100 and multiplied by the number of study year 1984–2023.

$$\text{Annual rate of change (\%)} = \frac{\text{Trend}}{\text{Number of study year}} \times 100 \tag{3}$$

The study relied on ground control points, Google Earth Engine (GEE), and high-resolution imagery as benchmark tools for validating classification. The classified images were compared with ground truthing data using confusion matrix, identifying the appropriate LULC class pixels using stratified random points in ArcGIS 10.8. The confusion matrix consist of overall accuracy (OA), producer’s accuracy (PA), user’s accuracy (UA), and kappa value. In addition, the overall accuracy showed the correctly signature pixels as reference, while Kappa coefficient indicate the agreement of occurrence by chance. The accuracy assessment is crucial in determining the reliability of LULC classes and effectiveness of the supervised classification algorithm (see Table 3). Validation samples were generated for each LULC classes using probability sampling design and statistical analysis. The formulas of OA was calculated using: $\frac{DP}{TRP}$, where DP is the total number of correctly classified diagonal pixels and TRP is the total number of referenced points.

$$K = \frac{Po - Pe}{1 - Pe} \tag{4}$$

where Po is the observed agreement and Pe is the expected agreement by chance, in consistent with hydrological modeling standard. An overall accuracy above 70% is considered an acceptable classification accuracy.

Flood inundation model analysis

This study adopted the Hydrologic Engineering Center’s Hydrologic Modeling System (HEC-HMS) for simulating rainfall-runoff processes and flood hydrograph estimations. Khan et al.²³ highlighted the key stages in the methodology including data input, model setup, calibration, validation, and scenario simulation. In this study, the Lagos watershed was assessed for possible flood damage in the HEC-HMS v4.2 software. Important flood damage indices including land extent, depth, and velocity to identify flood scenarios were considered⁷⁴. The pre-processed DEM and LULC, stream discharge, and rainfall data were used for the analysis⁴². Time-series data of a particular peak month was subjected to statistical average as part of the parameters for the HEC-HMS model. An analysis of terrain and drainage characteristics of the study area was presented in Fig. 3.

Delineation of sub-basins

The DEM was employed in basin and sub-basin creation using the hydrological tool in ArcMap 10.8 environment, extracting the flow direction grid, flow accumulation, stream order, and basin delineation. The DEM image was subjected to sinks fill in order to derive flow direction and accumulation. Basin delineation and sub-basins were generated for the watershed as recommended by Bagyaraj et al.⁷⁵.

Creation of curve number map

The study adopted Curve Number (CN) grid for each sub-basin by engaging in two steps. At first, potential runoff of land surface was prepared using a CN Lookup table. This contained various soil variations, land-use

LULC class	1984		2002		2023	
	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)
Built-up area	94	92	96	95	99	98
Forest	90	93	88	90	91	92
Light vegetation	85	88	89	91	94	96
Bareland	89	91	91	93	96	97
Waterbody	98	97	98	99	99	99
Wetland	86	84	90	88	95	96
Overall accuracy	0.92		0.94		0.97	
Kappa coefficient	0.90		0.92		0.96	

Table 3. Accuracy assessment outcomes for the land cover classification. PA: Producers Accuracy, UA: User’s Accuracy.

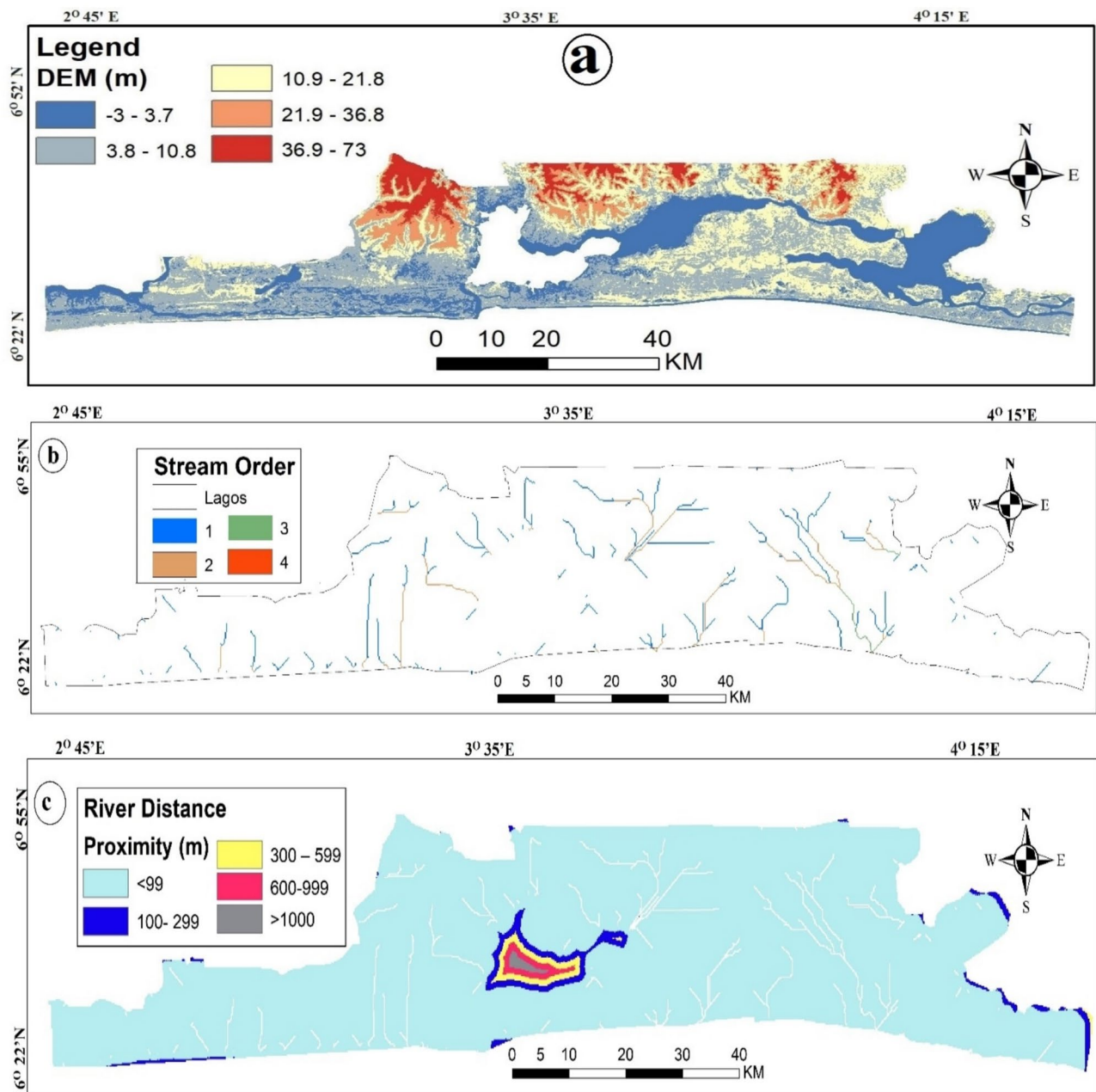


Fig. 3. Flood-conditioning factors of surface inundation.

groups, hydrological and moisture conditions that were defined for optimal validation using the Standard CN preconditions. Various land-use features that were identified within the Lagos basin were corresponded with Hydrologic Soil Group (HSG) of the area to determine their infiltration capacity in line with CN table values. By implication, a value of 100 indicates high potential runoff while a value of 0 indicates no runoff potential which corresponds with the rainfall variability of the study area. In addition, permeable soil has a higher rate of infiltration, while surface water/wetland corridors are assigned with a value of zero (0). However, a CN map was created using HEC-GeoMS consisting of sub-basins and CN for the area. While stream routing and water loss method was configured using SCS Curve Number (CN) method to estimate infiltration losses.

Model set-up, time lag calculation and validation

The study conducted rainfall and discharge, sub-basins quantification, and the reach extent. The model explained SCS-CN value variations based on the accumulated rainfall potentials exceeding the land use capacities, or otherwise, which is estimated using the following equation: $Pe = (P - Ia)^2 / (P - Ia + S)$; where, Pe represents the accumulated rainfall excess at time (mm); P is the accumulated rainfall depth per time (mm); Ia connotes initial abstraction (mm) = 0.2; and S is the potential maximum retention (mm). The calculation of lag time for the sub-

basin was dependent on the lag of peak runoff and loss method implemented in the Soil Conservation Service (SCS) approach.

Assessment of time lag considered the unit of hydrograph which is related to the time of concentration, peak runoff, and loss method in an ungauged watershed. The focus is on the total area (km²), stream length, and the topographic slope which was analyzed in the GIS environment for sub-basins. The study specified a meteorological model for rainfall and discharge hyetograph, connecting the sub-basins to the sink. The study applied Control Specifications from 01:00 h on 01-June-2023 to 12:00 h on 30-June-2023, with a time interval of one hour. The selection of 24th June 2023 was based on the highest mean value in the data, hence rainfall and discharge were linked to corresponding sub-basins in the study area. An hour time series data of rainfall (mm/h) and drainage discharge (m³/s) were calibrated into the model, preparing it for simulation run based on specified date and time.

The study followed Bowei et al.⁷⁶, using the following model parameters (CN, Tc, initial abstraction, & reach routing) on urban flooding simulations, validated through Model performance of Goodness-of-fit statistics of Nash–Sutcliffe Efficiency (NSE=0.79), Root Mean Square Error Deviation (RMSE=12.65 m³/s), Bias Percent (14.97%), and coefficient of determination (R²=0.80) for the study. The HEC-HMS model outputs were geospatially visualized using spatial analysis tool in ArcMap 10.8, revealing different time scenarios of flood.

Urban flood receptors prediction (Markov chain modeller)

The study considered anticipated changes in future LULC patterns based on urban growth dynamics. Therefore, the Markov chain modeller was engaged to predict future receptors of urban flood patterning. Markov chains determine a changing system of special classes in transitional probabilities ascribed to proportional matrix. Markov chains represent a dynamic system of special classes involving transition probabilities described as symmetric matrices⁶¹. This LULC prediction will quantify the projected flood vulnerability zones using classified LULC (vegetated and non-vegetated) in the matrix probability. Markov Chain Modeler utilized the classified images for TM 1984, ETH + 2002, and OLI 2023 for 2050 projection in the Terrset 18.31 software.

The LULC changes observed between 1984 and 2023 were the catalyst for processing the transitional matrix of change for the classified LULC classes, with expected transitional cell values obtained from the same periods. These transitional square probabilities for the six LULC classes formed a platform for predicting the LULC changes expected in 2050. The Markov Chains Modeler took into consideration the expected chances and complexity faults that might reduce the quality of the result by creating cell values for each of the classified images. With this, the consistency of the Markov Chain Modeler in predicting accurate LULC changes in Lagos Metropolis is achieved. The transition matrix generates its equation as a square matrix where each element *Pij*, *Pij* represents the probability of transitioning from LULC class changes *ii* to state *jj*. This is shown in the matrix table below:

$$P = \begin{matrix} & \begin{matrix} P_{11} & P_{12} & P_{13} \\ P_{21} & P_{22} & P_{23} \\ P_{31} & P_{32} & P_{33} \end{matrix} \end{matrix} \tag{5}$$

where *P11* is the probability of staying in LULC class 1 and *P12* represent the probability of moving from LULC class 1 to LULC class 2 ..., LULC class 6. The probability of the future transition depends on the current state multiplied by the sequence of previous states. This is mathematically expressed as: $P(X_{n+1} = x | X_n = x_n, X_{n-1} = x_{n-1}, \dots, X_0 = x_0) = P(X_{n+1} = x | X_n = x_n) P(X_n = x_n | X_{n-1} = x_{n-1}, \dots, X_0 = x_0) = P(X_{n+1} = x | X_n = x_n)$. The Markov Chain specified the validation using the training of image 1984–2002, validated with image 2023, with overall accuracy (87%) and Kappa coefficient (0.83).

Results

Land use and land cover dynamics between 1984 and 2023

The results show that the land use and land cover (LULC) of the study area have undergone continuous spatial changes over the past four decades, as revealed by the LULC classes and change detection analysis presented in Table 4 and visualized in Fig. 4. The total area of the study was 357,700.1hectares (ha) or 3770 square kilometers (km²), with the LULC classification examining changes in the years 1984, 2002, and 2023. The LULC was classified into six categories: built-up, forest, light vegetation, bareland, waterbody, and wetland. The changes in land area calculations and pattern of the overall landscape, detailing the types and extent of changes in the Lagos

LULC classes	1984		2002		2023	
	Hectares	%	Hectares	%	Hectares	%
Built-up area	55,403.82	15.49	91,651.68	25.62	190,202.2	53.17
Forest	166,499.82	46.55	55,772.01	15.59	49,150.37	13.74
Light vegetation	5.4	0.02	12,924.45	3.61	27,300.79	7.63
Bareland	4.5	0.01	53,055.45	14.83	9820.43	2.75
Waterbody	89,557.11	25.04	64,073.34	17.91	49,289.89	13.78
Wetland	46,229.49	12.92	80,223.21	22.43	31,936.39	8.93
Total	357,700.1	100.0	357,700.1	100.0	357,700.1	100.0

Table 4. Comparison of change in land use/land cover from 1984 to 2023.

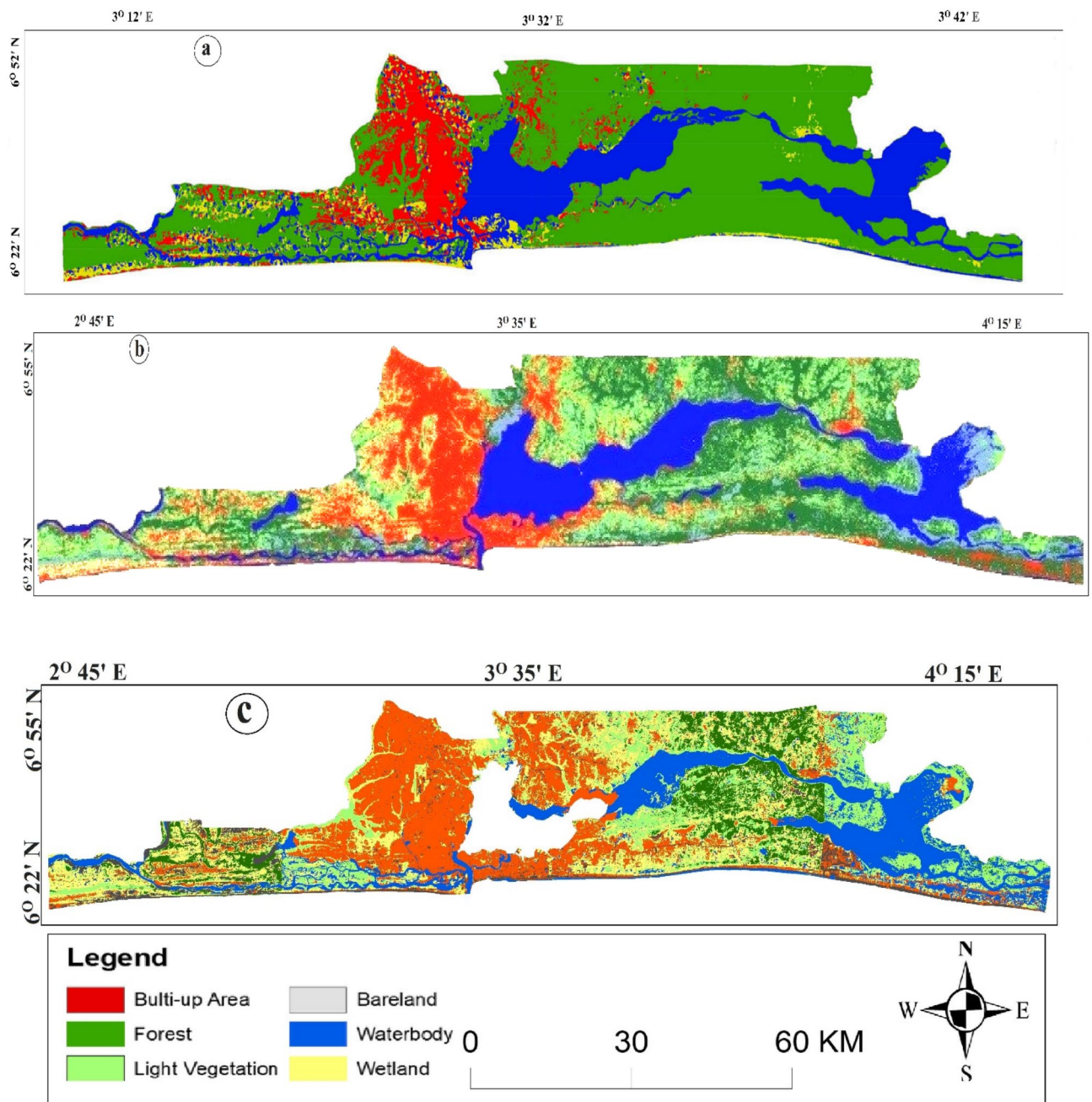


Fig. 4. (a–c): Land use Land Cover Classes of (a) 1984, (b) 2002, and (c) 2023.

Metropolis. In the years considered, forest (46.55%), waterbody (25.04%), built-up area (15.49%), and wetland (12.92%) showed significant percentages of each class in the total land area in 1984, while light vegetation (0.02%) and bareland (0.01%) were insignificant in that year.

In year 2002, built-up area (25.62%), wetland (22.43%), waterbody (17.91%), forest (15.59%), and bareland (14.83%) had significant percentages of their respective classes in the total land area, while light vegetation (3.66%) showed a minor significance in the same year. Moreover, the built-up area increased significantly to 53.17% as well as light vegetation (7.63%), while the forest declined to 13.74% in 2023. Likewise, changes were also observed in wetland (8.93%), waterbody (13.78%), and bareland (2.75%) in the study area in that year. These findings indicate that LULC changes from 1984 to 2023 have resulted in significant modifications to the landscape of the Lagos Metropolis, highlighting the direct impacts of the LULC changes on the flood characteristics and spatial distribution of urban floods.

Proportion of gains and losses of land use dynamics

The trend of LULC changes as shown in Table 5, revealed a significant loss in forest (– 30.95%) and waterbodies (– 7.12%), while bareland (14.83%), built-up areas (10.13%), wetland (9.50%), and light vegetation (3.66%)

LULC Classes	1984–2002		2002–2023		1984–2023	
	Hectares	$\Delta\%$	Hectares	$\Delta\%$	Hectares	$\Delta\%$
Built-up Area	36,247.86	10.13	98,550.52	27.55	134,798.38	37.68
Forest	–110,727.81	–30.95	–6621.64	–1.85	–117,349.45	–32.81
Light Vegetation	12,919.05	3.61	14,376.34	4.02	27,295.39	7.63
Bareland	53,050.95	14.83	–43,235.02	–12.09	9815.93	2.74
Waterbody	–25,483.77	–7.12	–14,783.45	–4.13	–40,267.22	–11.26
Wetland	33,993.72	9.50	–48,286.82	–13.50	–14,293.1	–4.00

Table 5. Proportion of land use cover gains and losses between 1984 and 2023. **Note:** Decrease in LULC class carries negative sign, $\Delta\%$ indicate percentage change in LULC.

increase between the years 1984 and 2002. Similarly, built-up areas (27.55%) and light vegetation (4.02%) increased from 2002 to 2023, while wetland (–13.50%), bareland (–12.09%), waterbodies (–4.13%), and forest (–1.85%) witnessed losses during that period. Throughout the entire study period from 1984 to 2023, built-up areas (37.68%) and light vegetation (7.63%) expanded significantly, at a rate of 3369.95 ha/yr^{–1} and 682.38 ha/yr^{–1} respectively. A minor increase was also observed in bareland (2.74%) over the years studied. On the other hand forest (–31.55%) and waterbodies (–11.26%) experience annual loss at the rate of 2933.74 ha/yr^{–1} and 1006.68 ha/yr^{–1} respectively across the study timeframe. Likewise, wetlands (4.32%) decreased from the original land area. The findings of the study revealed that as built-up areas and light vegetation increased, a corresponding decreases were noticeable in the other land cover classes. The changing patterns in LULC, as shown in Fig. 5 for the years 1984, 2002, and 2023, reflected a high rate of conversion within the LULC classes, particularly forest to built-up areas. Similarly, waterbodies witnessed substantial annual encroachment of 804.18 ha, indicating dredging and riverine conversion into residential areas that are highly vulnerable to floods. The first phase of urban transformation (1984–2002) was characterised by emerging flood zones through the conversion of wetland, loss of forests, and early urbanization. The second phase recognizes the period of frequent and severe flashfloods, where natural flood buffers have shrunk due to rapid urbanization and loss of urban greenery. The overall evaluation of urban flooding between 1984 and 2023 revealed a possible chronic and multi-hazard flooding, compounded by intensified LULC changes.

Flood inundation model analysis

The flood inundation assessment utilized DEM, LULC, rainfall, and stream flow data in the ArcMap v10.8 and HEC-HMS v4.12 software. The simulation was run in seven sub-basins using mean hydro-meteorological data. The streamflow discharge of September (peak month) and rainfall of June (peak month) create the flood inundation model. Table 6 displays the hydrological characteristics of flood inundation in the study area. Results showed that peak discharge varied based on parameters like soil type, impermeable surfaces, and basin capacity. Peak discharge (m³/s) was observed sub-basin 1 (12.5), sub-basin 2 (7.0), sub-basin 3 (4.2), sub-basin 4 (4.5), sub-basin 5 (0.4), sub-basin 6 (5.8), and sub-basin 7 (1.5). The average rainfall volume was 498 mm, covering a drainage area of 374.9 km².

Findings revealed that there is uniform specific discharge (0.36 m³/s/km²), indicating consistent rainfall intensity and similar land or soil conditions across the watershed. It was also observed that sub-basins exhibit very few variation, while consistent in volume (498 mm) across the sub-basins and reaches contributed nearly equal runoff depth, reinforcing that the storm was spatially uniform. Moreover, the sink behavior show a peak discharge (35.9 m³/s) and volume (498.50 mm) match closely with upstream inputs, suggesting a minimal losses or storage in the basin system direct runoff. The study highlighted differences in flood inundation characteristics, emphasizing the role of soil infiltration capacity, urbanization, and drainage network density. High rainfall amount and low discharge rates significantly impacted excess flow, leading to surface water inundation in all sub-basins examined.

Analysis of sub-basin hydrologic inundation

Table 7 presents the spatial distribution of peak discharge, rainfall, loss, excess and impermeability characteristics in each sub-basin examined for the study. Findings of the study establish noticeable excess flow in all the basins, as well as high rainfall and impervious surfaces within the region. The results revealed that surface water loss is significantly low compared to the amount of rainfall generated by the model, indicating increased inundation of the urban area. Noticeable peak discharge was observed in sub-basin 1 (12.5 m³/s), while sub-basin 5 (0.4 m³/s) had the lower discharge rate. Figure 6 reveal an increase in excess flow rate corresponding with the rainfall volume, indicating more land depth prone to flooding.

The flood depth increased significantly starting from the 5th to the 30th June, connoting a 25-day period of flooding resulting from rainfall intensity and excess water rate. Figure 4c revealed the sink capacity captured as low-lying, flat, and observed slow flow rate. However, simulated flow rises steadily to peak discharge, then falls on the 17th of June, indicating a very different behavior in a water reservoir. The reaches routing is highly limited or underestimated, leading to prolonged excess flow in the region. In addition, findings of the study revealed that RSME (12.65), Nash Sutcliffe (0.79), bias percent (14.97), and R-Squared (0.80) assessed the appropriateness and performance of the model. The highlighted results show satisfactory values that are generally acceptable.

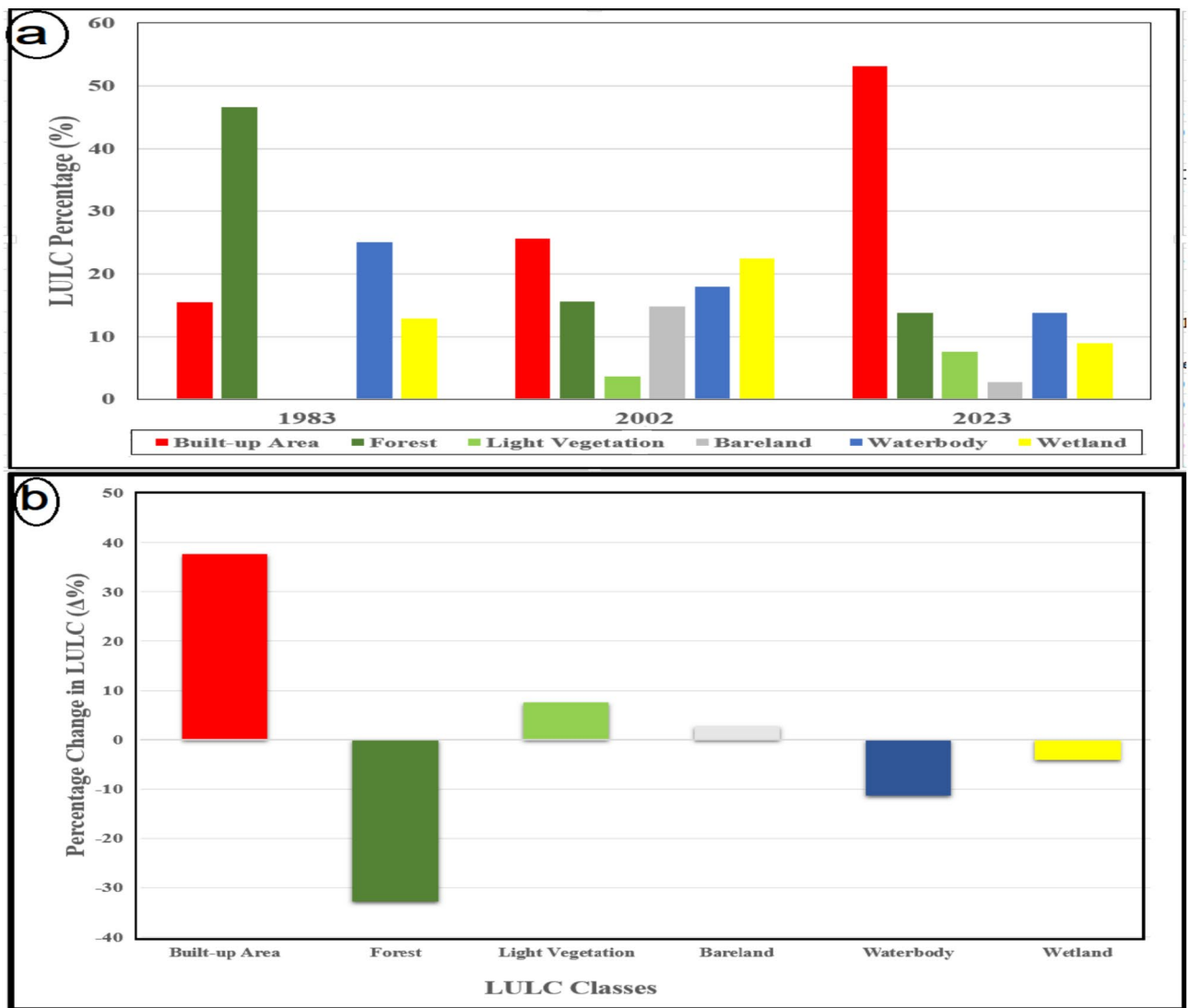


Fig. 5. (a–b): Land Cover Changes (a) Comparison among the Land-use and Land Cover Classes, and (b) Losses and Gains Analysis.

Hydrological element	Drainage area (km ²)	Peak discharge(m ³ /s)	Peak of time	Volume (mm)
Subbasin-1	34.8	12.5	24 June 2023, 01:00	498.17
Reach-1	95.9	34.4	24 June 2023, 01:00	498.47
Subbasin-2	19.5	7.0	24 June 2023, 01:00	498.01
Reach-2	54.4	19.5	24 June 2023, 01:00	498.11
Subbasin-3	11.7	4.2	24 June 2023, 01:00	499.43
Reach-3	24.2	8.7	24 June 2023, 01:00	498.75
Subbasin-4	12.5	4.5	24 June 2023, 01:00	498.12
Subbasin-5	1.1	0.4	24 June 2023, 01:00	498.24
Subbasin-6	16.3	5.8	24 June 2023, 01:00	499.33
Subbasin-7	4.3	1.5	24 June 2023, 01:00	499.16
Sink	100.2	35.9	24 June 2023, 01:00	498.50

Table 6. Hydrological characteristics of flood inundation modelling.

Sub-basin	Peak discharge (m3/s)	Rainfall (mm)	Loss (mm)	Excess (mm)	Impervious (%)
1	12.5	605.5	107.33	498.17	87.2
2	7	605.5	107	498.01	87.1
3	4.2	605.5	105.98	499.52	88.5
4	4.5	605.5	107.38	498.12	87.2
5	0.4	605.5	107.1	498.49	87.5
6	5.8	605.5	106.04	499.46	88.5
7	1.5	605.5	106.31	499.19	88.2
Computation point	RMSE	Nash sutcliffe	Bias (%)	R2	Remark
Sink	12.65	0.79	14.97	0.80	Acceptable

Table 7. Results of sub-basin hydrologic inundation and validation model.

The hydrograph analysis indicates a strong relationship between the simulated, observed runoff, and discharge stream flow as the major constituents of urban flooding in Lagos metropolis.

Spatial extent of flood inundations

This study presents findings on rainfall-runoff inundation and spatial flood extent across the Lagos Metropolis under three scenarios: low, moderate, and high flood extent. Results of the flood-prone zones were overlaid on land use types (vegetated and built-up areas) across all 20 LGAs in Lagos State, providing a comparative spatial visualization of increasing inundation severity as shown in Fig. 7. Table 8 provides a summary of flood inundation classification. The low flood model revealed a maximum land area to be affected is 46, 018.18 ha out of the total land area of 71, 552.6 ha, within a duration of less than 12 h. This scenario is typical of short-term floods, possibly resulting from localized rainfall and drainage issues. Though they may cover a wide area, the average impact is minimal and quickly recedes.

Similarly, the severity of flood based on the moderate area affected and temporal extent of inundation revealed that a maximum land area of 103, 665.78 ha out of the total land area of 125, 218.43 ha were affected, showing a mean land area of 20.15 ha was flooded with a duration of 24 h. The moderate flood scenario established a more widespread and longer-lasting event, indicating prolonged rainfall or overflow from water bodies occurrence. While the impact is moderate, it poses significant risks to infrastructure and mobility of people and vehicles.

Additionally, findings revealed that high flood inundation affected a maximum land area of 211, 230.22 ha out of 280, 578.01 ha of the total land area examined. It was observed that a mean affected land area is 86.4 ha during a 30-day duration assessment. This represents prolonged flood events, attributed to riverine or coastal flooding. The high mean value suggests severe impact in specific zones. Long-term displacement and damage are strongly associated with the high flood tendency of this type.

Extreme urban flooding and future flood receptors

Table 9 and Fig. 8 present the results of future flood receptors including the Markov chain analysis and the extent of extreme flooding for the year 2050. Findings from the predictive modelling revealed the expected urban flooding within the LULC classes for the projected year. The projected LULC focused on flood receptor categories such as built-up areas, forest, light vegetation, bareland, waterbodies, and wetlands. By the year 2050, an area of extreme flooding will significantly affect the built-up area, totaling 236,810.7 ha (66.20%), dominating the landscape and showing substantial urban expansion. Natural flood buffers including forest (1.35%), light vegetation (7.64%), and waterbodies (10.86%) declined drastically over the years, indicating noticeable environmental changes and high flood vulnerability in the future. Similarly, wetlands (11.18%) showed a decline in flood mitigation capability, while bareland (2.7%) revealed a pattern in land use change and possible flood risk.

However, the maximum affected area is 277, 403.96 ha (77.6% of the total land area), indicating a high potential for severe flooding, while the mean affected area of 300.40 ha demonstrates an average from multiple simulation runs or localized flood spots. The total affected area is 280, 578.01 ha (about 78.5% of the entire area), emphasizing widespread flood vulnerability. This severity of flooding based on the extreme area affected and future extent of inundation, reveals that a maximum land area of 277,403.96 ha out of the total land area of 280,578.01 ha will be affected by 2050. This simulated flood scenario predicts a more widespread and longer-lasting event on the urban landscape, affecting residents within the built-up areas.

The findings present results on predictive impact of extreme flood areas and population vulnerabilities within the major urban metropolitan area where flooding has socio-economic impacts. The spatial overlap of LASEMA flood reports reflect the high-flood impacts on buildings, roads, and linear structures in the areas, particularly projected land area subjected to extreme flood predictions. High risk LGAs which demonstrate the highest proportional interlinkage of population clumps with extreme flood risk areas include Eti-Osa, Ajeromi/Ifelodun, Lagos Island, Lagos Mainland, Surelere, Amuwo-Odofin, Ibeju-Lekki, Badagry, and Ojo.

Discussion

The present study analyzed land use dynamics and urban flood risk in Lagos megacity, modelling future flood scenarios and mitigation measures. It engaged multi-temporal remote sensing imageries and GIS obtained from 1984 to 2023, reflecting the past and current land use practices as an integer of future flood expectations.

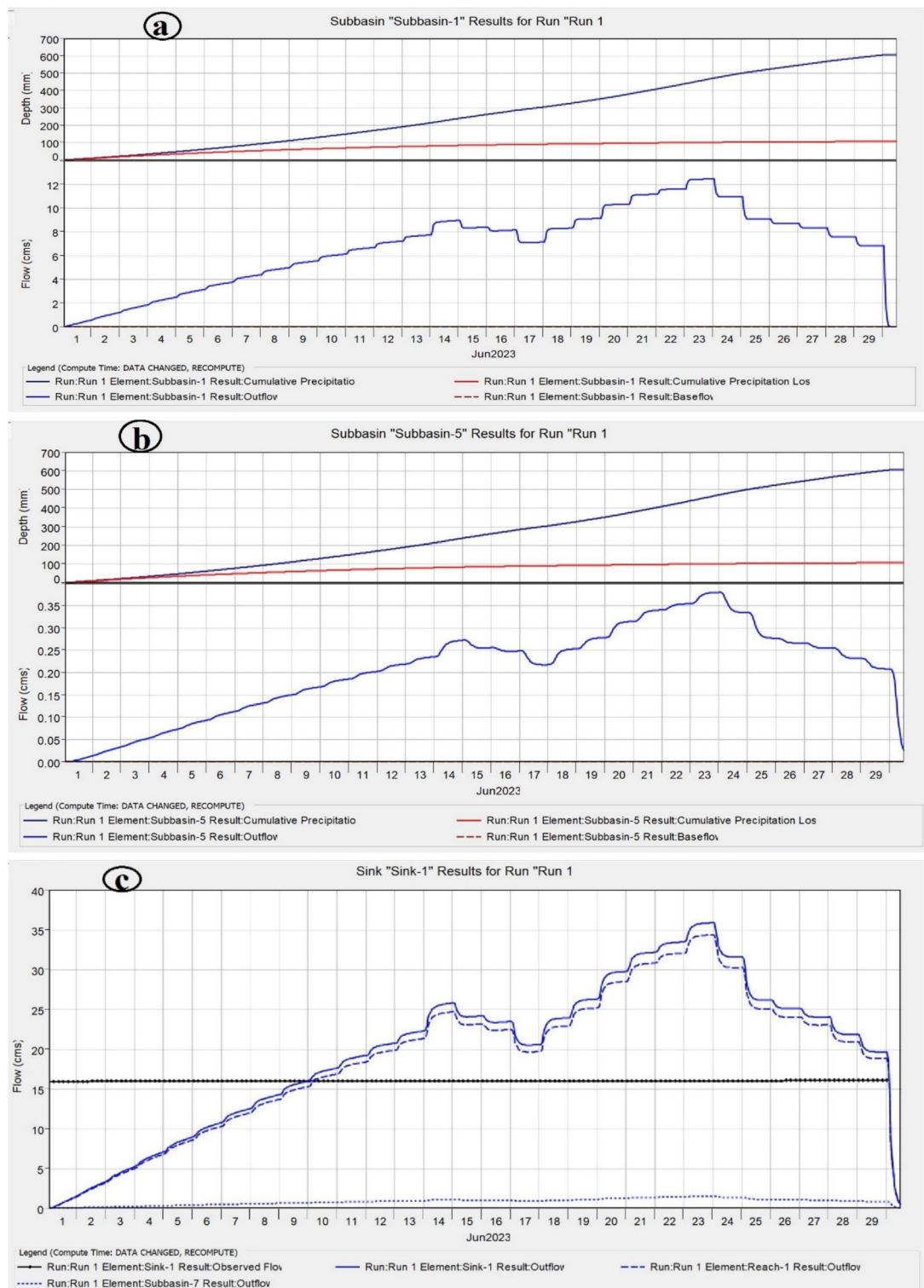


Fig. 6. (a–c) The simulated hydrograph of (a) sub-basin with highest peak, (b) sub-basin with lowest peak, and (c) outflow sink.

Understanding these typologies are essential for accurate flood risk assessment and urban planning, particularly in urban metropolitan areas where flood severity is greatly challenging. The scarcity of long-term hydrological records in Lagos megacity were integrated for mitigation framework in a typical megacity of a global south.

Studies by Arowolo and Deng⁷⁷, Chen et al.⁷¹, Idowu et al.³¹, and Arif et al.²⁷, agree with the development of LULC flood causes, rapidly increasing impermeable surfaces, burdening water drainage capacity, and groundwater intrusion. In this study, urban transformation from forest to built-up (14.54%) and light vegetation

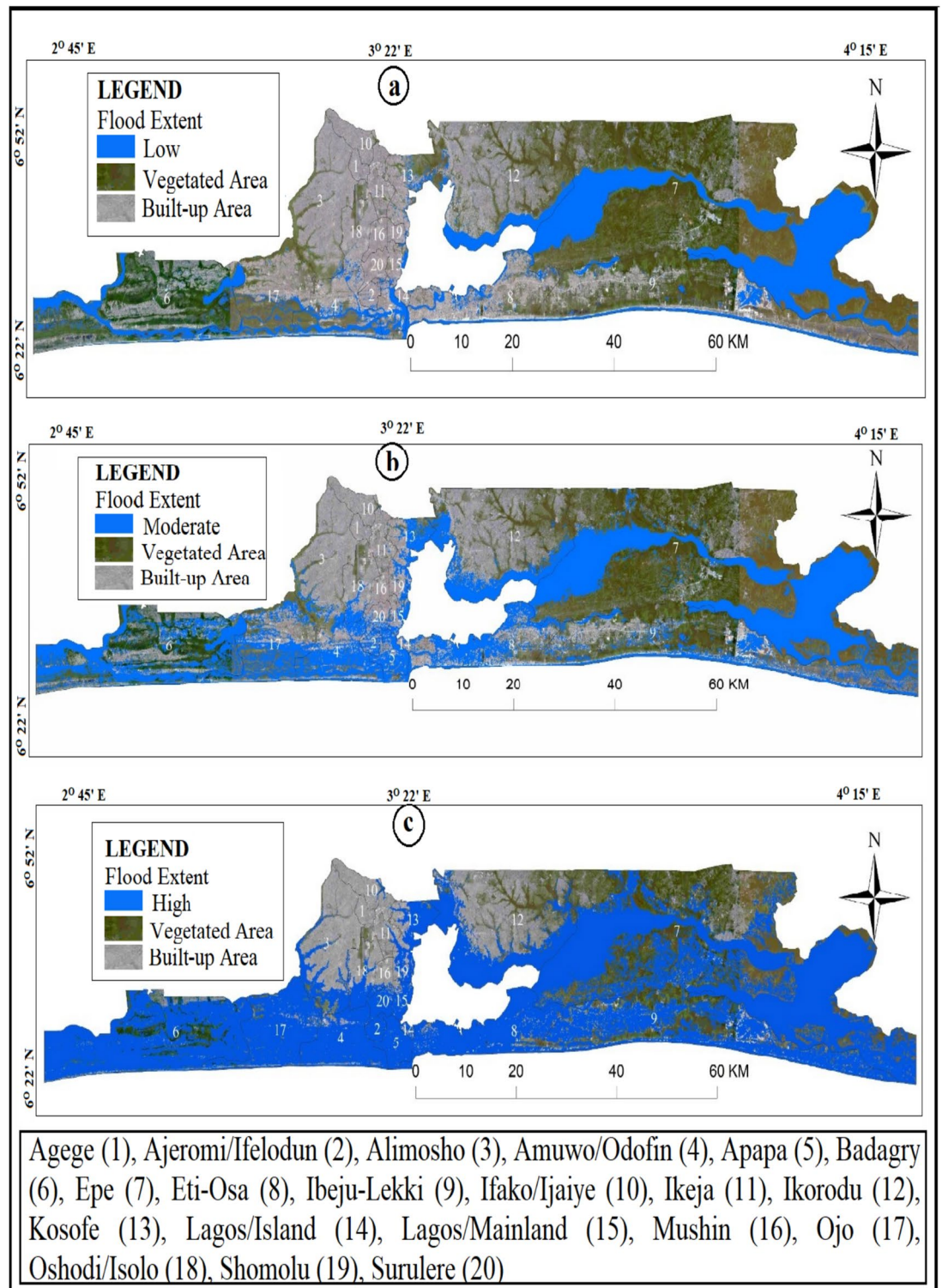


Fig. 7. (a–c) Aerial Extent of Floodwater Level in (a) Low Flood Level, (b) Moderate Flood Level, and (c) High Flood Level.

(18.87%) was significantly high, creating more urban vulnerability to flood problems. An in-depth examination into land use/land cover change detection revealed a forest cover loss of 31.55% in the total land area during the 40-year study. This created a significant increase in flood cases in the built-up area, establishing a major backdrop in implementing ecological solutions to urban flood challenges. Nkwunonwo et al.⁷⁸, Kaya and Derin⁷⁹, and Ogunwumi and Ihinegbu⁸⁰, validates flood impacts and urbanization effect in some countries in the global south. Hoque et al.⁸¹, Enoguanbhor et al.⁸², and Isiaka et al.⁵⁵ assessed the contributing factors of

Class	Maximum affected area (ha)	Mean affected area (ha)	Total area affected (ha)	Duration (Hour/day)
Low	46,018.18	17.18	71,552.6	< 12 h
Moderate	103,665.78	20.15	125,218.43	12–24 h
High	211,230.22	86.4	280,578.01	30 days

Table 8. Spatial and temporal flood inundation extent.

LULC classes	2050	
	Hectare	%
Built-up area	236,810.7	66.20
Forest	4820.43	1.35
Light vegetation	27,331.79	7.64
Bareland	9906.78	2.77
Waterbody	38,850.37	10.86
Wetland	39,979.98	11.18
Total	357,700.1	100.00

Extreme flood extent (ha)		
Maximum affected area	Mean affected area	Total area affected
277,403.96	300.40	280,578.01

Table 9. Area prediction of land-use change by 2050 and future flood extreme.

floods including poor urban layout, irregular settlement patterns, and poor channelization. The pattern of land use/land cover changes reflects the trend of urban flood dynamics, characterizing the vulnerability of residents over a period of four decades⁷⁶. Population growth responded to a high rate of forest conversion, influencing land management techniques and various aspects of ecological solutions to flooding. Alam et al.⁸³ highlights the significance of Nature-based Solutions (NbS) and urban greenery as an impetus to lower flood impact in highly vulnerable floodplains.

This continuous urban growth occurred in coastal and low-lying areas including Ikoyi, Ajegunle, Ojo, Lagos Island, and Mainland, experiencing unanticipated flooding over the years⁴. These findings align with Avashia and Garg⁸⁴, Bibi and Kara⁸⁵ and Olukunga et al.⁵⁴, stating that urbanization, low-impact development practices, and investment in real estate drastically lower the available ecological integrity of forests, wetlands, and water bodies to manage flooding, transforming natural flood buffers to residential, industrial, and economic gains. Evidently, urban development of Lagos Islands including popular locations such as Eko Atlantic City, Orange, Diamond, Parkview, Gracefield Phoenix Islands, Dangote Estate in Banana Island, and Petroleum Refinery in Ibeji-Lekki explained the rapid modification of land covers in vulnerable coastal regions, stretching towards the central area and northern axis of the study area (see Fig. 4). The findings of LULC 2023 showed a higher urban expansion (built-up 66.2%) and greater modeled flood-prone areas than previously reported by Faisal-Koko et al.⁵², Isiaka et al.⁵⁵. Sibandze et al.⁸⁶, while Aniramu and Orimoogunje,³⁸ emphasized that population density aligns with reported areas of constant flooding, unsustainable urban growth and enhancing resilience through SWOT-AHP approach.

Furthermore, urban flood characteristics are strongly embedded in hydro-climatic changes over time⁸⁷. The spatial dynamics modelling of rainfall and flood distributions has shown that there is moderate flood inundation in Lagos metropolis. Estimates of urban flood depth indicate a moderate and high flood discharge affecting an average land area of 20.15 and 86.4 hectares respectively (Table 8), estimating the variation in the duration of flood depth condition (Fig. 7). The floods induced by rainfall are mainly caused by heavy rainfall, particularly in June, July, and September, highlighting the poor infiltration capacity of the land and urban infrastructure. Findings of the study consonant with Hamdan et al.⁴², rainfall-runoff modelling using the HEC-HMS model had significant inundation impacts on the ecological resources. Built-up area is characterized by pavement and concrete, preventing water infiltration and intensifying surface inundation within the sub-basins of Lagos Metropolis.

In particular, the HEC-HMS simulated data align with finding of Hidayah et al.⁸⁸ reflecting how the shift in vegetated and wetland surfaces to impervious built-up areas in 2002–2023 led to the estimated increase in the peak runoff by 23–31%, which is dependent on the sub-basin attributes. The main factors that lead to such increase include decreased capacity of infiltration, decrease in lag time, and high values of Curve Number (CN) that is linked to urban growth. Such hydrological reactions have direct cause of spatial flood intensity, linking future LULC-driven CN changes with flood hazard intensification, which is rarely quantified for West African coastal megacities. The most significant impacts are seen in densely built-up or poorly planned urban areas such as Ajeromi/Ifelodun, Shomolu, Agege, Lagos Mainland/Island, and Isolo/Oshodi. Surface runoff and flash flooding also lead to damage to property, roads, and public infrastructure⁸⁹. These floods occurs seasonally, aligning with the intensity and frequency of hydro-climatic events, exacerbated by the rate of urbanization and unsustainable human practices⁹⁰. Additionally, backwash floods in Lagos UBD are worsened by a poor inland

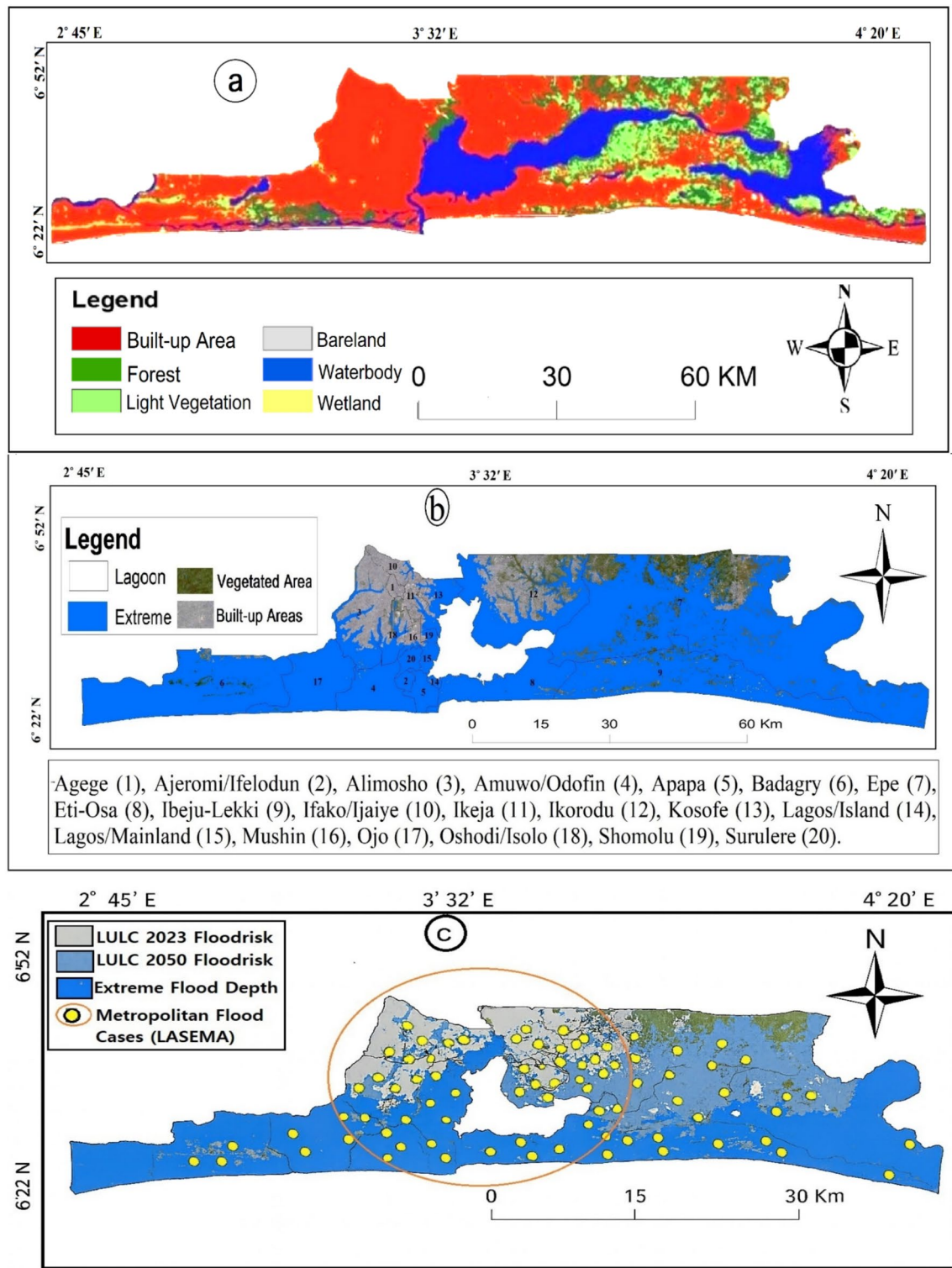


Fig. 8. (a–c) Future flood receptors (a) Land-use land cover of 2050, (b) Extreme flood extent, and (c) Intersection of land-use dynamics and flood episodes.

drainage system. The poorer populations often reside in low-income areas with substandard housing conditions and inadequate infrastructure, advancing their vulnerability to flood damage¹⁷. The study validates the works of Fashae et al.²¹, and Yoade et al.⁹¹, interconnected vulnerabilities and compounded flood risks attributed to the simultaneous effects of rainfall and backwash floods in many informal residents of Lagos coastal megacity and other flood zones in Nigeria. Siabi et al.⁹² reported that unsustainable urban development is a threat to socioeconomic pathways in West African megacities.

The severity of flood impact is exceptionally high in informal settlements where social vulnerability is characterized by poor standard housing, slum conditions, and inadequacies of basic amenities⁹³. The urbanization growth rate and population explosion estimated to be around 26 million people in year 2025 have recreated a national dilemma of flood impacts on people, businesses, and livelihood sustenance⁴. Existing urban flood facilities are highly incapacitated, urban planners' failure has promoted urban slums for decades, while flood consequences have become critical. Markov chain model and extreme flood cases were examined using the projected LULC of flood simulation, assessing the long-term effects on urban flood receptor in the future. Van-Berchum et al.⁹⁴, Kilsdonk et al.⁹⁵ observed that the effects of floods are linked to climate change and urban growth, which support the current findings that an average of 300 hectares of Lagos land area will be flooded in the future. Similarly, significant flood probability of high severity on urban population is predicted, noting the future flood receptors are accelerated by current practices of LULC transformation⁹⁶. As SDGs mandate of 2030 approaching, it is imperative to strategize plans against future flood challenges, addressing urban flood mitigations and ecological restoration. The current challenges of hydro-climatic flooding, rapid urbanization, and flood infrastructure inefficiency, call for a more transformative pathway towards a resilient future, accelerating the progress on the SDGs achievement while reducing flood risks⁵³.

Conclusion

Flooding in Lagos has become pervasive, prompting more proactive adaptation and community engagement. This highlights spatial flood risk across Lagos, revealing concentrated vulnerability within central metropolitan zone where built-up areas and land use dynamics are at the highest. The study provided a spatially assessment of future land use dynamics, reflecting its accompanying flood eventualities. By integrating geospatial techniques, hydro-climatic data, and predictive model, the study identified critical drivers of recurrent flooding, highlighting temporal and spatial patterns of flood occurrence, and evaluated future possibilities of flood receptors. The findings align with several Sustainable Development Goals (SDGs), particularly SDG 11 (Sustainable Cities and Communities), SDG 13 (Climate Action), and SDG 15 (Life on Land), emphasizing key gaps in urban planning, disaster risk reduction, and natural flood buffer conservation. The study established a strong linkage between urban growth and the decline in forest coverage within the examined region, converting natural flood buffers into vulnerable settlements (SDG 15). The continuous encroachment into ecologically sensitive areas has limited flood buffering, prolonging residential inundation during flash floods. This flood prevalence in coastal and high settlement areas reflects weak regulation and urban planning development in the study area, indicating inefficiencies of appropriate authorities in regulating environmental and economic development for the well-being of urban residents.

Implication of the study underscores the importance of implementing policies that are informed by spatial assessment of land utilization and urban landscape ecology. These policies should prioritize flood risk zoning, wetland conservation, and infrastructure investment in Lagos State, Nigeria. The study advocates mitigation measures on future vulnerability of flood, considering frequency and extreme climate conditions of Lagos growing population which is at high risk of flood severity. Therefore, the need for prevention efforts in Lagos through collaboration among flood management agencies and immediate framework to address the flood challenges in require now. Future research directions should advance academic knowledge and practical flood management solutions on the contributions of land elements such as roads, building clusters, and drainage infrastructures in flood zones. The integration of technical, social, and policy dimensions would provide holistic insights into sustainable land use, urban development, and disaster reduction efforts.

Limitation, uncertainty and direction of future study

We acknowledge the limitation arising from the use of observed streamflow data up to 2018 for simulating flood conditions projected to 2050. This temporal gap reflects the unavailability of long-term future discharge observations and is common in scenario-based hydrological assessments. In addition, the 30 m SRTM-DEM may pose micro-topographic variations which is critical for urban flood routing in flat coastal environments. We clarify that this limitation may introduce uncertainty in flood depth and localized inundation extents, and we emphasize that results should be interpreted at a strategic planning scale rather than parcel-level precision. However, historical streamflow was used strictly for model calibration and validation, ensuring that basin hydrological behavior is realistically represented. Future flood conditions were therefore simulated using projected land-use change and extreme rainfall scenarios rather than extrapolated discharge trends. While this approach supports robust scenario analysis, we recognize that incorporating climate-model-derived future discharge could further reduce uncertainty.

The present research admits the poor statistics on flood fatalities, in particular, compromised and politicized data in vulnerable flood communities, limiting the accuracy of socio-economic flood impacts and disasters reduction efforts in developing nations. Additionally, the Markov Chain model assumes stationary transition probabilities and does not capture non-linear drivers such as sudden policy changes, rapid economic development, or extreme climate events. Future studies should concentrate on interdisciplinary methods, working with stakeholders to address the effects of flooding in urban settings. Future resilience also depends on creating integrated hydrological model, socio-economic data, and non-linear drivers that incorporate specific land-use change scenarios in Lagos with high-resolution climate projections into predictive modeling.

Data availability

Raw data were generated at the United States Geological Survey (USGS) Earth Resources Observation and Science (EROS) Center ([<https://earthexplorer.usgs.gov/>] [<https://earthexplorer.usgs.gov/>]). Derived data supporting the findings of this study are available from the corresponding author based on request.

Received: 3 September 2025; Accepted: 29 January 2026

Published online: 10 February 2026

References

- Der Sarkissian, R., Al Sayah, M. J., Abdallah, C., Zaninetti, J. M. & Nedjai, R. Land use planning to reduce flood risk: Opportunities, challenges and uncertainties in developing countries. *Sensors* **22**(18), 6957 (2022).
- Idowu, D. & Zhou, W. Land use and land cover change assessment in the context of flood hazard in Lagos State. *Nigeria. Water* **13**, 1105 (2021).
- Zhang, H., Jia, H., Li, C. & Zhang, Q. Exploring the driving factors of urban flood at the catchment scale: A case study of multitype megacities in China. *Ecol. Indic.* **166**, 112513 (2024).
- Onajomo, T. S. Geospatial Assessment of Land Use/Cover, Rainfall, and Flood Incidents in Eti-Osa, Lagos, Nigeria. *S. Afr. J. Geomat.* **11**(2), 202–218 (2022).
- Adegboyega, S., Oloukoi, J., Olajuyigbe, A. & Ajibade, O. Evaluation of unsustainable land use/land cover change on ecosystem services in coastal area of Lagos state, Nigeria. *Appl. Geomat.* **11**(1), 97–110 (2019).
- Aniramu, O. S., Adetoro, O. O. & Salami, A. T. Implication of human induced activities on ecotourism in Ikogosi warm spring centre, Ekiti State, Southern western, Nigeria. *J. Geoinform. FCE CTU* **17**(1), 33–44 (2018).
- Luo, Z., Tian, J., Zeng, J. & Pilla, F. Resilient landscape pattern for reducing coastal flood susceptibility. *Sci. Total Environ.* **856**, 159087 (2023).
- United Nations-Habitat (2024) World Report: Mapping the Solution Space for Climate Action: The Role of Urban Planning and Design. Available online: https://unhabitat.org/sites/default/files/2024/11/wcr_2024_-_statistical_annex.pdf (Accessed: May 5, 2025)
- United Nations Department of Economic and Social Affairs (2018) World Urbanization Prospects Highlights. Available Online: <https://population.un.org/wup/assets/WUP2018-Report.pdf>. (Accessed: May 5, 2025)
- World Bank Group (2020) Urban Population-Nigeria. World Urbanization Prospects: 2018 revision. Available online: <https://data.worldbank.org/indicator/SP.URB.TOTL.IN.ZS?locations=NG>. (Accessed: May 5, 2025)
- Lagos Sustainability Summit (2024). Highlights from the Lagos State Sustainability Summit: A Vision for a Resilient Lagos. Available online: <https://www.csr-in-action.com/highlights-from-the-lagos-state-sustainability-summit-a-vision-for-a-resilient-lagos/> (Accessed: January 11, 2025).
- Youssef, A., Sewilam, H. & Khadr, Z. Impact of Urban sprawl on agriculture lands in Greater Cairo. *J. Urban Plan. Dev.* **146**, 05020027 (2020).
- Obiefuna, J., Adeaga, O., Omojola, A., Atagbaza, A. & Okolie, C. Flood risks to urban development on a coastal barrier landscape of Lekki Peninsula in Lagos, Nigeria. *Sci. Afr.* **12**(2021), 1–17 (2021).
- Adeola, O., Aniramu, O., Adenegan, O. & Orimoogunje, O. Land use change, planning instruments, and sustainable strategies for urban flood risk management in Ibadan, Nigeria. *Discov. Environ.* **3**, 284. <https://doi.org/10.1007/s44274-025-00478-2> (2025).
- Adededeji, O. H. Analysis of landscape pattern bases on the CA-markov model. *Appl. Geogr.* **112**, 301–318 (2022).
- Lawanson, O. I., Proverbs, D. & Ibrahim, R. L. The impact of flooding on poor communities in Lagos State, Nigeria: The case of the Makoko urban settlement. *J. Flood Risk Manag.* **16**(1), 1–16 (2023).
- Shao, Z. et al. Urban sprawl and its impact on sustainable urban development: A combination of remote sensing and social media data. *Geo-Spat. Inf. Sci.* **24**, 241–255 (2021).
- Tobore, A., Senjobi, B. & Oyerinde, G. Spatiotemporal analysis and simulation pattern of land use and land cover change in Odeda Peri-urban of Ogun State, Nigeria. *Jordan J. Earth Environ. Sci.* **12**(4), 326–336 (2021).
- Roy, P. et al. Threats of climate and land use change on future flood susceptibility. *J. Clean. Prod.* **272**, 122757 (2020).
- Abdelkareem, M. & Mansour, A. M. Risk assessment and management of vulnerable areas to flash flood hazards in arid regions using remote sensing and GIS-based knowledge-driven techniques. *Nat. Hazards* **117**(3), 2269–2295 (2023).
- Fashae, O. A. et al. Comparative assessment of the changing pattern of land cover along the Southwestern coast of Nigeria using GIS and remote sensing techniques. *Sci. Afr.* **17**(6), e01286 (2022).
- Ijaware, V. A. GIS-based flood vulnerability studies for Ife Central Local Government Area, Osun State, Nigeria. *Niger. J. Environ. Sci. Technol.* **4**, 160–171 (2020).
- Khan, I. R., Elmahdy, S. I., Rustum, R., Khan, Q. & Mohamed, M. M. Floods modelling and analysis for Dubai using HEC-HMS model and remote sensing using GIS. *Sci. Rep.* **14**, 25039 (2024).
- Thapa, P. Using geospatial technologies for disaster management in developing countries. *J. Remote Sens. GIS* **9**(4), 13 (2021).
- Zope, P., Eldho, T. & Jothiprakash, V. Impacts of land use–land cover change and urbanization on flooding: A case study of Oshiwara River Basin in Mumbai, India. *CATENA* **145**, 142–154 (2016).
- Onilude, O. O. & Vaz, E. Data analysis of land use change and urban and rural impacts in Lagos state, Nigeria. *Sci. Data* **5**, 72 (2020).
- Arif, M., Sengupta, S., Mohinuddin, S. K. & Gupta, K. Dynamics of land use and land cover change in Peri-urban area of Burdwan City, India: A remote sensing and GIS-based approach. *GeoJournal* **88**, 4189–4213 (2023).
- Gowtham, S. & Dhanapriya, M. Land use land cover (LULC) change detection of east & south eastern coastal plain of Odisha from 1990–2018 and its impact over land surface temperature (LST). *J. Exp. Agric. Int.* **47**(6), 1–26 (2025).
- Idowu, T. E., Waswa, R. M., Lasisi, K., Nyadawa, M. & Okumu, V. Object-based land use/land cover change detection of a coastal city using multi-source imagery: A case study of Lagos, Nigeria. *S. Afr. J. Geomat.* **9**, 136–148 (2022).
- Ige-Olumide, O. O. & Salami, A. T. Implication of land use change on flood vulnerability in an urban watershed of Ibadan, Southwest, Nigeria. *J. Geograph. Environ. Earth Sci. Int.* **14**(1), 1–17 (2018).
- Ajibade, F. O. et al. Flood-prone area mapping using GIS-based analytical hierarchy frameworks for Ibadan city, Nigeria. *J. Multi-Criteria Decis. Anal.* **28**, 283–295 (2021).
- Akinyemi, F. O. Land change in the central Albertine rift: Insights from analysis and mapping of land use-land cover change in north-western Rwanda. *Appl. Geogr.* **87**, 127–138 (2017).
- Chinedu, A. D., Ezebube, N. M., Uchegbu, S. & Ozorme, V. A. Integrated assessment of flood susceptibility and exposure rate in the lower Niger Basin, Onitsha, Southeastern Nigeria. *Front. Earth Sci.* **12**, 1394256. <https://doi.org/10.3389/feart.2024.1394256> (2024).
- Ozezin, K. O. & Ilugbo, S. O. Evaluation of potentially susceptible flooding areas leveraging geospatial technology with multi-criteria decision analysis in Edo state, Nigeria. *Nat. Hazards Res.* <https://doi.org/10.1016/j.nhres.07.002> (2024).
- Rahman, M. et al. Flooding and its relationship with land cover change, population growth, and road density. *Geosci. Front.* **12**, 101224 (2021).
- Shuaibu, A., Hounkpè, J., Bossa, Y. A. & Kalin, R. M. Flood risk assessment and mapping in the Hadejia River Basin, Nigeria, using hydro-geomorphic approach and multi-criterion decision-making method. *Water* **14**(22), 3709 (2022).
- Adetunji, M. & Oyeleye, O. Evaluation of the causes and effects of flood in Apete, Ido Local government Area, Oyo state, Nigeria. *J. Flood Risk Manag.* **4**(1), 25–44 (2020).
- Aniramu, O. & Orimoogunje, O. Integrating geospatial mapping and stakeholders' perception on sustainable flood solutions in a typical coastal Megacity of Nigeria: A SWOT-AHP approach. *Front. Sustain. Cities* **7**, 1663269. <https://doi.org/10.3389/frsc.2025.1663269> (2025).

39. Aslan, V. Evaluation of Birecik Basin flood detection with GIS-assisted AHP methods. *KSCE J. Civ. Eng.* **27**(7), 2891–2907 (2023).
40. Das, S. Flood susceptibility mapping of the Western Ghat coastal belt using multi-source geospatial data and analytical hierarchy process (AHP). *Remote Sens. Appl. Soc. Environ.* **20**, 100379 (2020).
41. Ewunetu, A., Simane, B. & Teferi, E. Mapping and quantifying comprehensive land degradation status using spatial multi-criteria evaluation technique in the headwaters area of upper Blue Nile River. *Sustainability* **13**(4), 2244 (2021).
42. Hamdan, A. N. A., Almuktar, S. & Scholz, M. Rainfall-runoff modeling using the HEC-HMS model for the Al-Adhaim River catchment, Northern Iraq. *Hydrology* **8**, 58–62 (2021).
43. Khouz, A. et al. Flood susceptibility assessment through statistical models and HEC-RAS analysis for sustainable management in Essaouira Province, Morocco. *Geosciences* **13**, 382 (2023).
44. Rangari V. A., Sridhar V., Umamahesh N. V. & Patel A. K. 2020 Rainfall Runoff Modelling of Urban Area Using HEC-HMS: A Case Study of Hyderabad City. In *Lecture Notes in Civil Engineering*. **39**. (Springer, 2020). https://doi.org/10.1007/978-981-13-8181-2_9.
45. Oyatayo, K. T., Ndabula, C., Jeb, D. N., Adamu, G. K. & Jidauna, G. G. Integrating digital elevation model, landuse/landcover and flood frequency analysis: A deterministic approach to flood inundation and risk modeling of Makurdi along Its River Benue Reach. *Fudma J. Sci.* **5**(1), 477–489 (2021).
46. Rangari, V. A., Bhatt, C. M. & Umamahesh, N. V. Rapid assessment of the October 2020 Hyderabad urban flood and risk analysis using geospatial data. *Curr. Sci.* **120**(12), 1840–1847 (2021).
47. Rentschler, J. et al. Global evidence of rapid urban growth in flood zones since 1985. *Nature* **622**, 87–92 (2023).
48. Salami, W. A., Tayo-Badru, M. A., Adeofun, C. O. & Oguntok, O. Effects of urban expansion on vegetation cover loss in Lagos state, Nigeria. *Afr. J. Sci. Nat.* **10**, 146–163 (2020).
49. Kasim, O. F., Wahab, B. & Oweniwe, M. F. Urban expansion and enhanced flood risk in Africa: The example of Lagos. *Environ. Hazards* **21**(2), 137–158 (2022).
50. Shen, Y., Morsy, M. M., Huxley, C., Tahvildari, N. & Goodall, J. L. Flood risk assessment and increased resilience for coastal urban watersheds under the combined impact of storm tide and heavy rainfall. *J. Hydrol.* **579**, 124159 (2019).
51. Lagos State Emergency Management Agency (LASEMA). Flood Vulnerability Assessment and Mapping of Lagos State using GIS Technique. (2023), available online at: chrome-extension://efaidnbmninnibpcapcglcfeindmkaj/<https://ludi.org.ng/wp-content/> (accessed on 21/07/2024).
52. Faisal Koko, A., Yue, W., AbdullahiAbubakar, G., Hamed, R. & Noman Alabsi, A. A. Analyzing urban growth and land cover change scenario in Lagos, Nigeria using multi-temporal remote sensing data and GIS to mitigate flooding. *Geomat. Nat. Hazards Risk* **12**, 631–652 (2021).
53. Orimoogunje, O. & Aniramu, O. systematic review of flood resilience strategies in Lagos metropolis: Pathways toward the 2030 sustainable development agenda. *Front. Clim.* **7**, 1603798 (2025).
54. Olukunga, O. A., Adewumi, B. J. & Hassan, T. Comparative analysis of flood mitigation strategies for residential housing in Lagos State. *African J. Environ. Sci. Renew. Energy* **15**, 209–225 (2024).
55. Isiaka, I. O., Gafar, S., Ajadi, S. A. & Mukaila, I. Flood susceptibility assessment of Lagos State, Nigeria using geographical information system (GIS)-based frequency ratio model. *Int. J. Environ. Geoinform.* **10**, 76–89 (2023).
56. Wang, J. & Maduako, I. N. Spatio-temporal urban growth dynamics of Lagos Metropolitan Region of Nigeria based on Hybrid methods for LULC modelling and prediction. *Eur. J. Remote Sens.* **51**(1), 251–265 (2018).
57. Nigeria Hydrological Service Agency (NIHSA) (2024) Annual Flood Outlook (Flood History, Mitigation, and Adaptation Measures). Available online: <https://nihsa.gov.ng/2024/04/18/honourable-minister-of-water-resources-unveils-2024-annual-flood-outlook-emphasizing-data-analytics-and-modeling-for-flood-risk-assessment-and-food-security/> (accessed: July 28th, 2025)
58. United States Geological Survey (USGS) (2020) Landsat Collection Datasets. Available Online: <https://earthexplorer.usgs.gov>. (Accessed: June 25 2024).
59. Hanzl, M. Monitoring urbanization and environmental impact in Kigali, Rwanda using Sentinel-2 MSI data and ecosystem service bundles. *Int. J. Appl. Earth Obs. Geoinf.* **109**, 137–144 (2021).
60. Kandissounon, G. A., Karla, A. & Ahmad, S. Integrating system dynamics and remote sensing to estimate future water usage and average surface Runoff in Lagos, Nigeria. *Civ. Eng. J.* **4**(2), 378–393 (2018).
61. Kumar, M. S. & Umamahesh, N. V. Integrated assessment of future climate and land use changes on urban floods: A Markov chain and PCSWMM-based approach for Hyderabad case study. *Water Sci. Technol.* **89**(4), 1003–1027 (2024).
62. Manchikatla, S. K. & Umamahesh, N. V. Simulation of flood hazard, prioritization of critical sub-catchments, and resilience study in an urban setting using PCSWMM: A case study. *Water Policy* **24**(8), 1247–1268 (2022).
63. Onilude, O. O. & Vaz, E. Urban sprawl and growth prediction for Lagos using GlobeLand30 data and cellular automata model. *Geo-Spatial Inf. Sci.* **3**, 23 (2021).
64. Kamaraj, M. & Rangarajan, S. Predicting the future land use and land cover changes for Bhavani basin, Tamil Nadu, India, using QGIS MOLUSCE plugin. *Environ. Sci. Pollut. Res.* **29**, 86337–86348 (2022).
65. Löschnner, L., Seher, W., Nordbeck, R., & Kopf, M. (2019). Blauzone Rheintal: A regional planning instrument for future-oriented flood management in a dynamic risk environment. *Nature-Based Flood Risk Management on Private Land: Disciplinary Perspectives on a Multidisciplinary Challenge*, 141–154.
66. Chamling, M. & Bera, B. Spatio-temporal patterns of land use/land cover change in the Bhutan—Bengal foothill region between 1987 and 2019: Study towards geospatial applications and policy making. *Earth Systems and Environment* **3**, 34–49 (2020).
67. Aneasha, S. B., Shashi, M. & Deva, P. Future land use land cover scenario simulation using open source GIS for the city of Warangal, Telangana, India. *Appl. Geomat.* **12**, 281–290 (2020).
68. Amiri, A., Soltani, K., Ebtehaj, I. & Bonakdari, H. A novel machine learning tool for current and future flood susceptibility mapping by integrating remote sensing and geographic information systems. *J. Hydrol.* **632**, 130936 (2024).
69. Tewabe, D. & Fentahun, T. Assessing Land Use and Land Cover Change Detection using Remote Sensing in the Lake Tana Basin, Northwest Ethiopia. *Cogent Environ. Sci.* **6**(1), 45–71 (2020).
70. Jensen, J. R. *Introductory Digital Image Processing: A Remote Sensing Perspective* 2nd edn. (Prentice Hall Inc, 1996).
71. Chen, Y., Ye, Z., Liu, H., Chen, R. & Liu, Z. A GIS-based approach for flood risk zoning by combining social vulnerability and flood susceptibility: A case study of Nanjing, China. *Int. J. Environ. Res. Public Health* **18**, 11597 (2021).
72. Anderson, J. R., Hardy, E. E., Roach, J. T. & Witmer, R. E. A Land Use and Land Cover Classification System for Use with Remote Sensor Data. Geological Survey Professional Paper No. 964, U.S. (Government Printing Office, 1976).
73. Bunyangha, J., Majaliwa, M. J. G., Muthumbi, A. W., Gichuki, N. N. & Egeru, A. Past and future land use/land cover changes from multi-temporal Landsat imagery in Mpologoma catchment, eastern Uganda. *Egypt. J. Remote Sens. Space Sci.* **24**, 675–685 (2021).
74. Cao, W. et al. Increasing global urban exposure to flooding: An analysis of long-term annual dynamics. *Sci. Total Environ.* **817**, 153012 (2022).
75. Bagyaraj, M. et al. A geospatial approach for assessing urban flood risk zones in Chennai, Tamil Nadu, India. *Environ. Sci. Pollut. Res.* **30**, 100562–100575 (2023).
76. Bowei, Z., Huang, G. & Wenjie, C. Research progress and prospects of urban flooding simulation: From traditional numerical models to deep learning approaches. *Environ. Model. Softw.* **183**, 106213 (2024).
77. Arowolo, A. O. & Deng, X. Land use/land cover change and statistical modelling of cultivated land change drivers in Nigeria. *Reg. Environ. Change* **18**, 247–259 (2018).
78. Nkwunonwo, U., Whitworth, M. & Baily, B. A review of the current status of flood modelling for urban flood risk management in the developing countries. *Sci. Afr.* **7**, e00269 (2020).

79. Kaya, C. M. & Derin, L. Parameters and methods used in flood susceptibility mapping: A review. *J. Water Clim. Change* **14**, 1935–1960 (2023).
80. Ogunwumi, T. and Ihinegbu, C. (2025) Last-mile household preparedness for future disasters: A study of flooding in Kogi State, Nigeria. *Afr. Geograph. Rev.* 1–20.
81. Hoque, M., Tasfia, S., Ahmed, N. & Pradhan, B. Assessing Spatial flood vulnerability at Kalapara Upazila in Bangladesh using an analytic hierarchy process. *Sensors* **19**, 130–142 (2019).
82. Enoguanbhor, E. C., Gollnow, F., Walker, B. B., Nielsen, J. O. & Lakes, T. Key challenges for land use planning and its environmental assessments in the Abuja City-Region. *Nigeria. Land* **10**(5), 443 (2021).
83. Alam, N., Saha, S., Gupta, S. & Chatterjee, A. Settlement Suitability analysis of a riverine floodplain in the perspective of GIS-based multi-criteria decision analysis. *Environ. Sci. Pollut. Res.* **30**(24), 66002–66020 (2023).
84. Avashia, V. & Garg, A. Implications of land use transitions and climate change on local flooding in urban areas: An assessment of 42 Indian cities. *Land Use Policy* **95**, 104571 (2020).
85. Bibi, T. S. & Kara, K. G. Evaluation of climate change, urbanization, and low-impact development practices on urban flooding. *Heliyon* **9**, e12955 (2023).
86. Sibandze, P., Kalumba, M., AbayomiAfuye, G. & Leocadia, Z. Geospatial mapping and meteorological flood risk assessment: A global research trend analysis. *Environ. Manag.* **15**, 137–154 (2024).
87. Porebska, A., Muszyński, K., Godyn, I. & Racón-Leja, K. City and water risk: Accumulated runoff mapping analysis as a tool for sustainable land use planning. *Land* **12**(7), 1345 (2023).
88. Hidayah, E., Widiarti, W. Y., Wiyono, R. U. A. & Lee, W. K. Nature-based solutions for flood risk reduction in the Bedadung River. *J. Water Clim. Change* **16**(10), 2883–2900 (2025).
89. Odunsi, O. M. et al. Households' Resilience to flood disaster in Lagos State, Nigeria: Developing a conceptual framework unifying disaster resilience components and dimensions. *J. Environ. Stud. Sci.* **14**, 69–86 (2024).
90. United Nations Office for Disaster Reduction (2023). Uncounted Costs of Disaster: Gaps Hide the True Human Impacts of Disasters in 2023. Available online at: <https://www.undrr.org/explainer/uncounted-costs-of-disasters-2023> (Accessed January 11, 2024).
91. Yoade, A. O., Onifade, V. O., Olatunji, S. A. & Husseni, A. Perception of flood risk coping and adaptive strategies of residents: Example of Mushin, Nigeria. *J. Surv. Constr. Prop.* **14**, 55–70 (2023).
92. Siabi, E. K. et al. Future land use simulation modeling for sustainable urban development under the shared socioeconomic pathways in West African megacities: Insights from Greater Accra Region. *J. Environ. Manag.* **376**, 124300 (2025).
93. Ologunorisa, T. E., Obioma, O. & Eludoyin, A. O. Urban flood event and associated damage in the Benue Valley, Nigeria. *Nat. Hazards* **111**, 261–282 (2022).
94. Van-Berchum, E. C., van-Ledden, M., Timmermans, J. S., Kwakkel, J. H., & Jonkman, S. N. Rapid flood risk screening model for compound flood events in Beira, Mozambique. *Nat. Hazards Earth Syst. Sci.* **20**, 2633–2646, (2020).
95. Kilsdonk, R. A., Bomers, A. & Wijnberg, K. M. Predicting urban flooding due to extreme precipitation using a long short-term memory neural network. *Hydrology* **9**, 105 (2022).
96. Nkeki, F. N., Bello, E. I. & Agbaje, I. G. Flood risk mapping and urban infrastructural susceptibility assessment using a GIS and analytic hierarchical raster fusion approach in the Ona River Basin, Nigeria. *Int. J. Dis. Risk Reduct.* **77**, 103097 (2022).

Acknowledgements

The study acknowledges the late Pastor Nathaniel Feso Aniramu for his suggestions and guidance in undertaking this research. Similar appreciation is extended to the Ekiti State Scholarship Board for their fellowship award on the doctoral dissertation, from which the manuscript is carved out.

Author contributions

Conceptualization, O.A., O.I. and O.O.; methodology, O.A., and O.O.; software and validation, O.A., O.I. and O.O.; formal analysis, O.A.; investigation, O.A., and O.O.; manuscript writing, O.A. and O.I.; supervision, O.O.; editing and review, O. A., O. I. and O.O.; All authors have read and agreed to the published version of the manuscript.

Funding

The study was not funded by government or non-governmental organization.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to O.A.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026